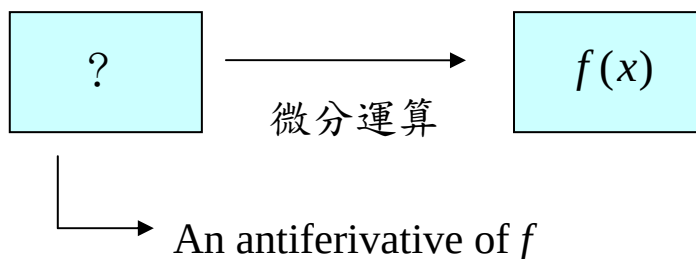


§4-10 Antiderivatives

Homework : 7,11,27,35,45,59,74.



Definition : If $F'(x) = f(x)$ for all $x \in I$, then F is called an antiderivative of f on I .

Example 1 : $f(x) = x$, $F(x) = \frac{x^2}{2} + 1$, $F(x) = \frac{x^2}{2} - 1$

Theorem : If F is an antiderivative of f on I , then the most general antiderivative of f on I is $F(x) + C$, where C is an arbitrary constant.

Example 2 : $f'(u) = \frac{u^4 + 3\sqrt{u}}{u^2}$, 求 $f(u)$.

Solution : $f'(u) = u^2 + 3u^{-\frac{1}{2}} \Rightarrow f(u) = \frac{u^3}{3} - 6u^{\frac{1}{2}} + C$.

Example 3 : $f'(x) = \sqrt{x}(6+5x)$, $f(1)=10$, 求 $f(x)$.

Solution : $f'(x) = 6x^{\frac{1}{2}} + 5x^{\frac{3}{2}}$

$$\Rightarrow f(x) = 4x^{\frac{3}{2}} + 2x^{\frac{5}{2}} + C$$

$$\Rightarrow f(1) = 4 + 2 + C = 10$$

$$\Rightarrow C = 4$$

$$\Rightarrow f(x) = 4x^{\frac{3}{2}} + 2x^{\frac{5}{2}} + 4.$$

Example 4 : A car traveling at 50 mile/hr when the brakes are fully applied, producing a constant deceleration of 22 ft/s². What is the

distance covered before the car comes to a stop ?

Solution : $a = -22 \text{ ft/s}^2$ $1 \text{ mile} = 5280 \text{ ft}$

$$V(t) = -22t + C, \quad V(0) = 50 \text{ miles/h} = \frac{220}{3} \text{ ft/sec.}$$

$$\Rightarrow V(t) = -22t + \frac{220}{3} \quad \text{令 } V(t) = 0 \Rightarrow t = \frac{10}{3}$$

$$\Rightarrow S(t) = -11t^2 + \frac{220}{3}t + C_1$$

$$d = S\left(\frac{10}{3}\right) - S(0) = \frac{1100}{9} \text{ (ft/sec)}.$$